

3.3 Unit 3 INFO3 The Use of ICT in the Digital World

This unit looks at the fast changing subject of ICT, including developments in technology and ICT system capabilities, and how this might affect the world that makes use of ICT.

The content is designed to address issues associated with the management of ICT and its use within organisations. A range of organisations should be studied, including charities, clubs and societies, small, medium and large businesses, both national and multi-national public and private organisations. The number of people involved could range from one or two to many thousands.

Each organisation will have the same ICT-related issues to consider on a scale appropriate to their

own needs, for instance, the introduction and testing of ICT systems, the training and support needed for users of ICT systems, the outside organisations which affect the way the organisation runs its ICT.

It should be remembered when teaching this unit that ICT systems are there to support the activities of organisations, and are essential in the modern world. It is expected that students will study much of the content of this unit through the investigation of examples of the uses made of ICT by organisations.

The Teacher Resource Bank will provide further amplification and a glossary of terms.

Topic	Key Concepts	Content and Amplification
1. Future developments	Emerging technologies.	Enabling devices for remote and mobile working. Advances in technology that give business and leisure benefits.
	Potential future uses of ICT.	Developments in uses of current and future ICT systems.
	Implications of future developments and future use of ICT.	The impact on society. The impact on the way organisations are run. The impact on individuals as consumers and as workers. The social, cultural, legal, technical, ethical, economic and environmental issues surrounding the rapid development of ICT.
2. Information and systems	Different organisations have different information needs.	Type and scale of organisation. Nature and management style.
	Different activities within an organisation have different information needs.	Ordering systems, customer support
	Different levels of task have different information needs.	Strategic, Tactical, Operational Managing, Operating
	Different personnel have different information needs.	Suppliers, customers, official and legal bodies.
	Organisations have to exchange information with external bodies.	Privacy, security, and legal compliance implications.
	ICT systems support the activities of organisations.	The role of ICT is to improve the efficiency and effectiveness of business processes.
	Common ICT systems exist such as payroll, personnel and accounting. There is a need for information to be transferred between ICT systems both internally and externally.	The demands of the organisation itself. The requirements of external agencies. Supply chain. New systems interfacing with legacy systems. See the Teacher Resource Bank for further guidance.

Topic	Key Concepts	Content and Amplification
	Types of ICT system and their uses.	Back Office Systems. Transaction processing systems; workflow and document management systems; system for collaborative working. Management Information Systems (MIS). Enterprise systems, Customer Relationship Management. Decision Support Systems (DSS). E-commerce. For example: online sales, ticket reservations, licence applications. See the Teacher Resource Bank for further guidance.
3. Managing ICT	The size of an organisation affects the degree of formality with which ICT is managed.	Small organisations have informal systems. Large organisations tend to have formal ones.
	An ICT strategy should match the long-term aims of an organisation. ICT policies outline how a strategy would be put into operation.	For example, an aim of the organisation to increase its market share might necessitate the ICT strategy, including the development of e-commerce.
	Contribution of ICT management to business strategy.	Chief Information Officer's role as member of the company executive.
4. ICT strategy	Factors that influence an ICT system strategy within an organisation.	For example, business goals, available finance, legacy systems, geography of clients and business fulfilment, external factors: compliance, legislation.
	The management of information assets over time.	Students need to be aware that the long-term use of ICT systems involves the need to manage an increasing volume of data.
	The need for a corporate strategy covering technology for ICT systems in large organisations.	Future proofing, developments in technology, procurement. Technology lifecycle, information management, people considerations.
	Standards exist that may affect strategic choices.	For example, for exchanging data.
5. ICT Policies	Organisations will have policies covering different aspects of the use of ICT, such as security, training and procurement.	
6. Legislation	The implication of legislation on ICT policies. Legislation will impact on procedures within an organisation.	

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7. Developing ICT solutions	Factors that contribute to a successful development process.	For example, Management and End User involvement at appropriate times; effective ICT teamwork.
	Factors during the development process that might contribute to the failure of a newly introduced system.	For example, inadequate analysis; losing control of the project plan.
8. Development methods	ICT systems need to be developed in stages.	Systems development life cycle.
	The stages of development.	Analysis of the problem and proposed solution. Design and specification of the solution. Constructing the solution (writing code, customising a package). Testing (Module, Functional, Systems, User, Operational). Installation and Conversion (including documentation, training and handover). Review and maintenance.
	The need for systematic, formal methods.	Project management; agreed deliverables, milestones, sign-off and approval to proceed.
	Development methodologies.	Students need to be aware of a variety of linear and iterative methods.
9. Techniques and tools for systems development	Investigating and recording techniques.	Students need to be aware of the techniques available and their appropriate uses.
	Business process modelling tools.	Students need to know, understand and be able to use diagrams and tools used in the modelling of business processes.
	Data modelling tools. Techniques for testing.	Students need to know, understand and be able to use diagrams and tools used in the modelling of data. Students need to be aware of the range of techniques available. For example, techniques such as test harnesses, volume testing, scalability, prototyping, multi-platform, use of simulated environments. See the Teacher Resource Bank for further advice.
10. Introducing large ICT systems into organisations	Scale	Large ICT systems can be used not only within a single organisation but also across multiple small organisations. For example, ordering systems used in newsagents; NHS systems used across the country in trusts but also in a single GP surgery.
	Reliability and testing.	Students need to be aware of the issues involved in testing large ICT systems: - ensuring that large ICT systems always operate as expected - designing testing to ensure reliable operation – techniques and the people involved - the specialist skills and facilities required for the testing of network based systems.

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	Installation	Essential features of a successful installation plan for a large-scale ICT system. Methods of introducing systems. Hardware installation and testing. Software installation and testing. Documentation. Resources (cross-referenced to training)
	Backup and recovery.	Strategies for backup of large scale systems. Risk Analysis. Scale of backup. Procedures for recovery of large scale systems. Disaster Recovery/Contingency planning. Options for recovery. Cross-reference to training.
	Maintenance	Types of maintenance on large scale systems. Adaptive Corrective Perfective Maintenance teams; handover. Cross-reference to User support.
11. Training and supporting users	Users of ICT systems can be both internal and external to the organisation. The training and support requirements may be different.	Different levels of staff need to learn different functionality of systems to match job and role requirements. External users may require training.
	Training	Training methods available. The relative merits of the different methods of training for: - the organisation - the individual being trained.
	Support	Support options available. Factors an organisation would consider when selecting a suitable support option.
	Customers	Interfaces available. The relationship between the choice of interface and business activities. Managing the interface between the organisation and its customers.

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12. External and internal resources	Using external ICT services and business support.	Students need to be aware of a range of ICT services available such as outsourcing, offshore, bulk printing (payroll, billing). Students need to be aware of different ways that organisations can obtain ICT services from suppliers such as contracting (people, space, equipment), leasing (software, communication links, equipment).
	Managing internal resources.	Planning the management and control of: • hardware resources • software resources • communication resources • consumables • facilities and power • people.